

# OLA Ride Booking Analytics – End-to-End Data Analytics Project

## Executive Summary

This project analyzes **103,024 OLA ride-booking records** to identify customer behavior, booking trends, cancellation patterns, revenue drivers, and service quality metrics. The solution was developed using **SQL, Excel, and Power BI** to transform raw operational data into actionable business intelligence.

The analysis revealed that OLA generated **₹35.08 Million** in revenue from successful rides, achieved a **62.09% ride completion rate**, and identified significant opportunities to reduce cancellations and improve customer experience.

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## Business Problem

OLA operates in a highly competitive ride-hailing industry where customer satisfaction and operational efficiency directly impact revenue.

Management wanted to understand:

- Why rides are being cancelled
  - Which vehicle categories drive revenue
  - Customer payment preferences
  - Ride completion performance
  - Customer and driver satisfaction levels
  - Opportunities to improve operational efficiency
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## Project Objectives

### Operational Objectives

- Analyze more than **100,000 booking transactions**
  - Measure ride completion and cancellation performance
  - Identify high-demand vehicle categories
  - Evaluate customer and driver satisfaction
  - Analyze payment behavior
  - Generate actionable business recommendations
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## Dataset Overview

| Metric             | Value          |
|--------------------|----------------|
| Total Records      | 103,024        |
| Vehicle Categories | 7              |
| Customer Records   | 100,000+       |
| Payment Methods    | 4              |
| Total Revenue      | ₹35.08 Million |

## Data Preparation

The dataset was cleaned and transformed using Excel before loading into SQL and Power BI.

### Cleaning Activities

- Removed duplicate entries
- Standardized booking status values
- Handled missing ratings
- Validated revenue and distance fields
- Verified vehicle category information

## Key Performance Indicators (KPIs)

### Booking Performance

| KPI                    | Value   |
|------------------------|---------|
| Total Bookings         | 103,024 |
| Successful Bookings    | 63,967  |
| Ride Completion Rate   | 62.09%  |
| Total Cancellations    | 28,933  |
| Cancellation Rate      | 28.08%  |
| Driver Not Found Cases | 10,124  |

## Insight

Nearly **1 out of every 3 rides failed to complete**, indicating a significant opportunity to improve driver allocation and ride fulfillment.

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## Revenue Analysis

| KPI                                 | Value       |
|-------------------------------------|-------------|
| Total Revenue                       | ₹35,080,467 |
| Average Revenue per Successful Ride | ₹548.42     |

## Revenue by Vehicle Type

| Vehicle Type | Revenue       |
|--------------|---------------|
| Prime Sedan  | ₹5.22 Million |
| eBike        | ₹5.05 Million |
| Auto         | ₹5.05 Million |
| Prime Plus   | ₹5.02 Million |
| Bike         | ₹4.97 Million |
| Mini         | ₹4.89 Million |
| Prime SUV    | ₹4.88 Million |

## Insight

**Prime Sedan** emerged as the highest revenue-generating category, contributing over **₹5.22 Million**.

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## Customer Analysis

### Payment Method Preference

| Payment Method | Transactions |
|----------------|--------------|
| Cash           | 35,022       |
| UPI            | 25,881       |
| Credit Card    | 2,435        |

| Payment Method | Transactions |
|----------------|--------------|
| Debit Card     | 629          |

### Insight

Cash remains the dominant payment mode, while UPI has strong adoption, indicating opportunities for digital payment promotions.

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## Customer Satisfaction

| Metric                  | Rating   |
|-------------------------|----------|
| Average Customer Rating | 4.00 / 5 |
| Average Driver Rating   | 4.00 / 5 |

### Insight

Customer satisfaction remains consistently high across all vehicle categories.

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## Driver & Operational Analysis

### Booking Status Distribution

| Status                | Count  |
|-----------------------|--------|
| Success               | 63,967 |
| Cancelled by Driver   | 18,434 |
| Cancelled by Customer | 10,499 |
| Driver Not Found      | 10,124 |

### Key Finding

Driver-related issues account for a significant proportion of ride failures:

- Driver cancellations: **18,434 rides**
- Driver unavailable: **10,124 rides**

Together they impact over **28,500 bookings**, highlighting operational inefficiencies.

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# Ride Distance Analysis

| KPI                   | Value    |
|-----------------------|----------|
| Average Ride Distance | 14.19 km |

## Insight

Most rides fall within the medium-distance category, indicating a strong urban commuting customer base.

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# Incomplete Ride Analysis

## Top Incomplete Ride Reasons

| Reason                 | Count |
|------------------------|-------|
| Customer Demand Issues | 1,601 |
| Vehicle Breakdown      | 1,591 |
| Other Issues           | 734   |

## Insight

Vehicle breakdowns and customer demand-related interruptions contribute significantly to service disruptions.

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# SQL Analysis Performed

The project involved developing multiple SQL queries to answer business questions:

## Queries Included

- Successful ride analysis
- Average distance by vehicle type
- Customer cancellation analysis
- Driver cancellation analysis
- Top 5 customers by booking frequency
- Revenue calculation
- Customer rating analysis
- Driver rating analysis
- Payment method analysis
- Incomplete ride investigation

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# Power BI Dashboard

The dashboard was developed using multiple analytical pages covering:

## Dashboard Pages

1. Overall Performance Dashboard
2. Vehicle Performance Dashboard
3. Revenue Analysis Dashboard
4. Cancellation Analysis Dashboard
5. Customer & Driver Ratings Dashboard

## KPIs Visualized

- Total Bookings
- Completion Rate
- Cancellation Rate
- Revenue
- Vehicle Performance
- Customer Satisfaction
- Payment Analysis

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# Business Recommendations

## Reduce Driver-Related Ride Failures

Since over **28,500 rides** were impacted by driver cancellations or unavailability:

- Improve driver availability forecasting
- Introduce peak-hour driver incentives
- Optimize driver allocation algorithms

## Increase Digital Payments

Although UPI adoption is strong:

- Offer cashback rewards for UPI payments
- Reduce dependency on cash transactions

## Improve Vehicle Reliability

Since vehicle breakdowns account for many incomplete rides:

- Implement preventive maintenance schedules

- Monitor vehicle health indicators

### Focus on Premium Segments

Prime Sedan generates the highest revenue:

- Expand premium ride offerings
  - Introduce loyalty programs for premium users
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## Project Impact

This analytics solution enabled:

- ✓ Analysis of **103,024 ride records**
  - ✓ Monitoring of **₹35.08 Million revenue**
  - ✓ Identification of **28,933 cancellations**
  - ✓ Tracking of customer and driver satisfaction
  - ✓ Development of an interactive Power BI dashboard
  - ✓ Data-driven recommendations to improve operational efficiency
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## Skills Demonstrated

### Technical Skills

- SQL
- Excel
- Power BI
- Data Cleaning
- Data Modeling
- Data Visualization
- KPI Development

### Business Skills

- Customer Analytics
- Revenue Analytics
- Operational Analytics
- Dashboard Design

- Business Intelligence
  - Data Storytelling
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## Conclusion

The OLA Ride Booking Analytics Project successfully transformed over **103,000 ride-booking records** into actionable business insights. Through SQL analysis and Power BI visualization, the project identified key revenue drivers, cancellation patterns, customer preferences, and operational improvement opportunities, helping stakeholders make informed business decisions.